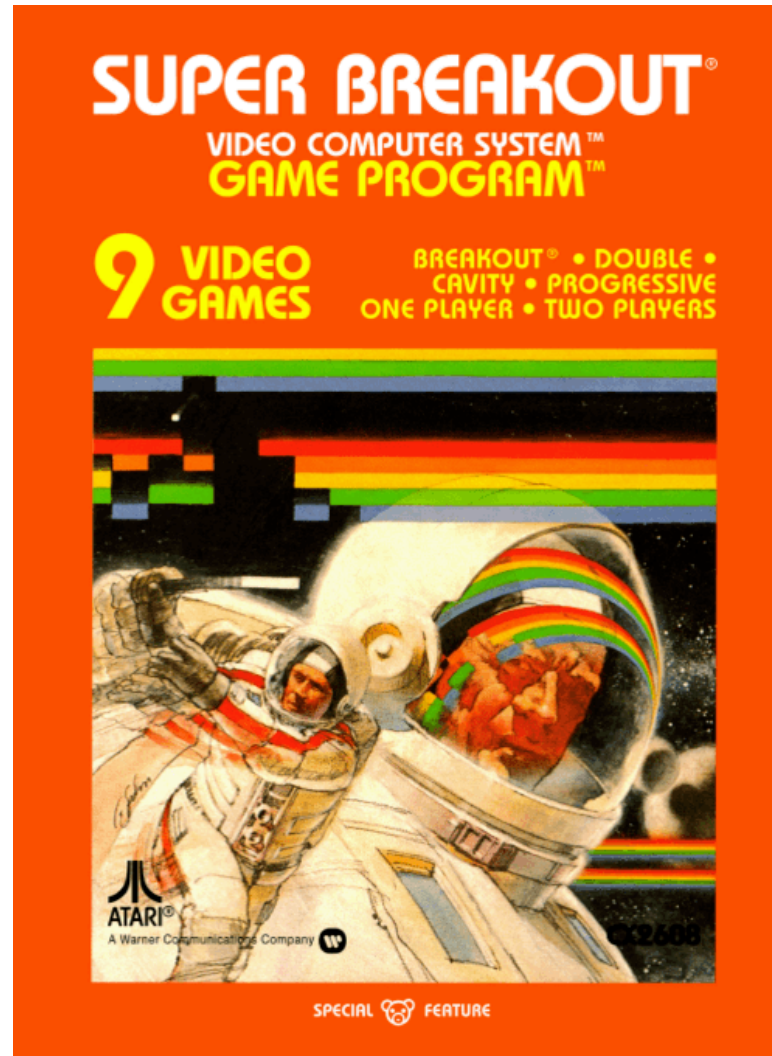


RL for POMDPs

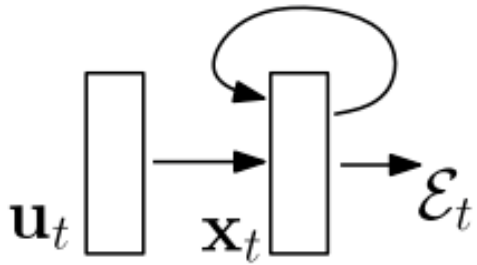
4 Approaches

Approach 1: k-Markov



Approach 2: Recurrent Neural Network

e.g. Alpha Star



Input: u_t

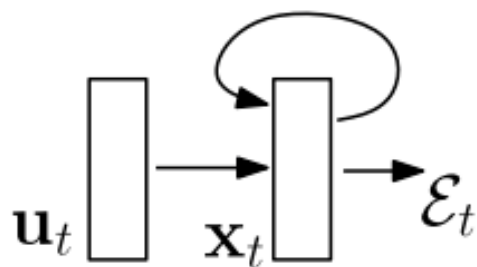
State/Output: x_t

Cost: $\mathcal{E}_t$

Images: Pascanu 2013

Approach 2: Recurrent Neural Network

e.g. Alpha Star

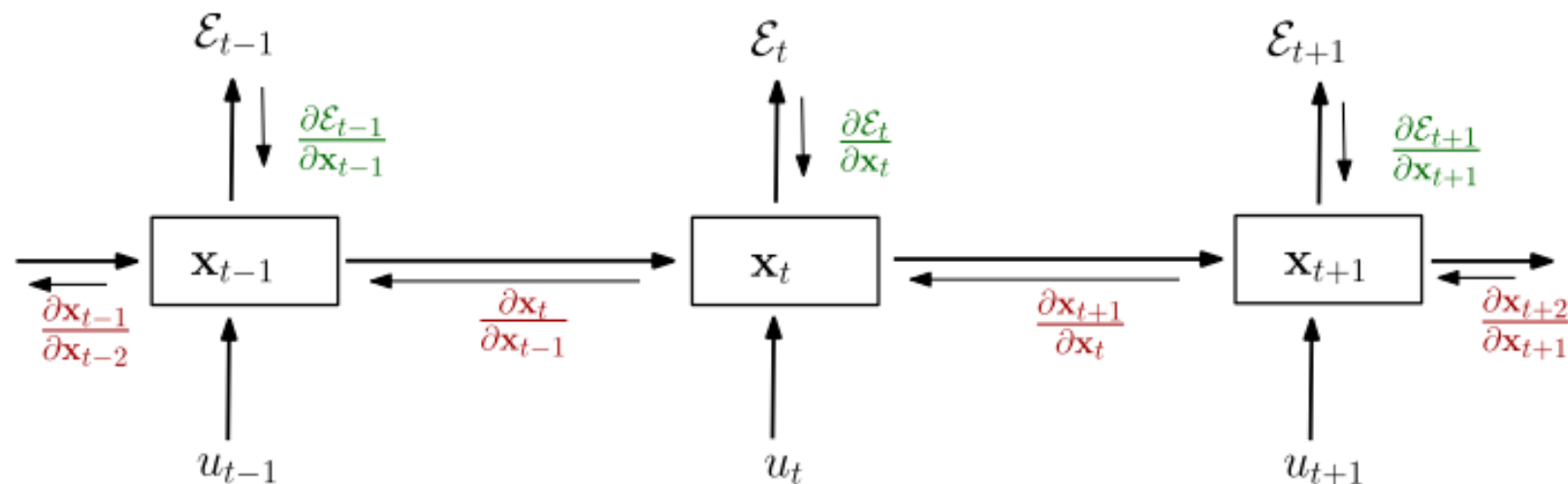


Input: u_t

State/Output: x_t

Cost: $\mathcal{E}_t$

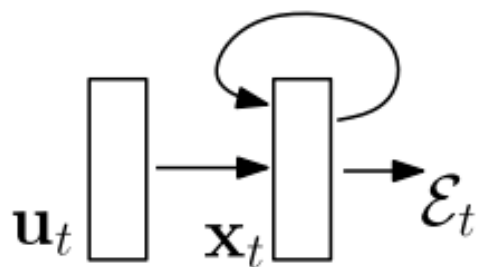
Training: Unrolling through time



Images: Pascanu 2013

Approach 2: Recurrent Neural Network

e.g. Alpha Star

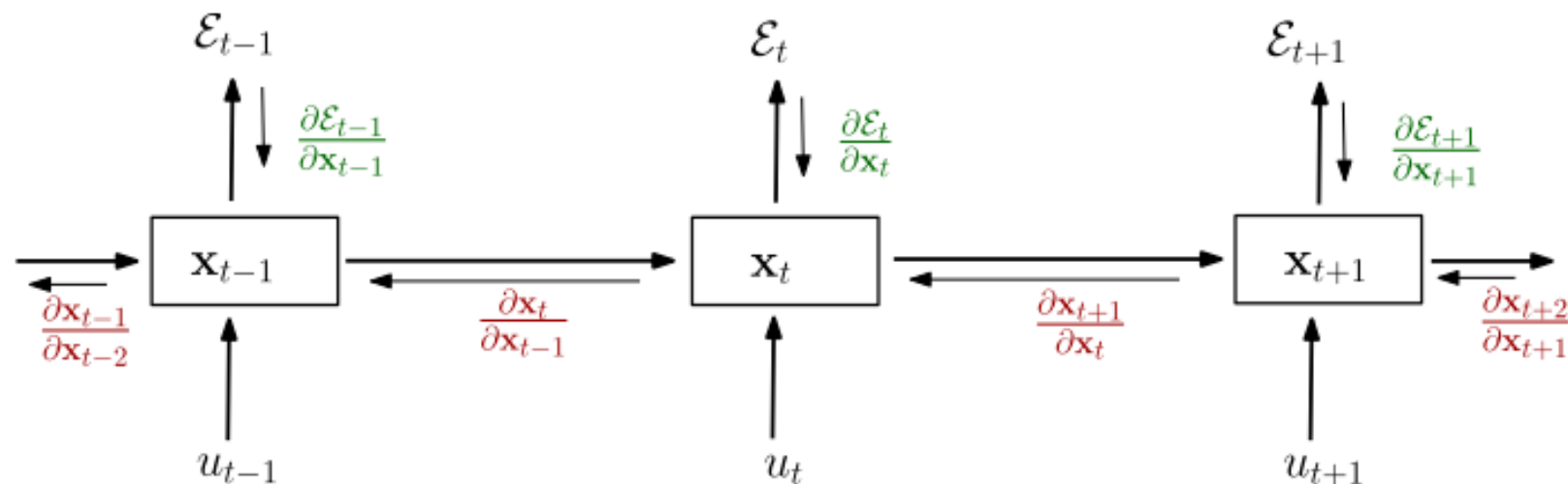


Input: u_t

State/Output: x_t

Cost: $\mathcal{E}_t$

Training: Unrolling through time



Problem: Exploding/Vanishing Gradients

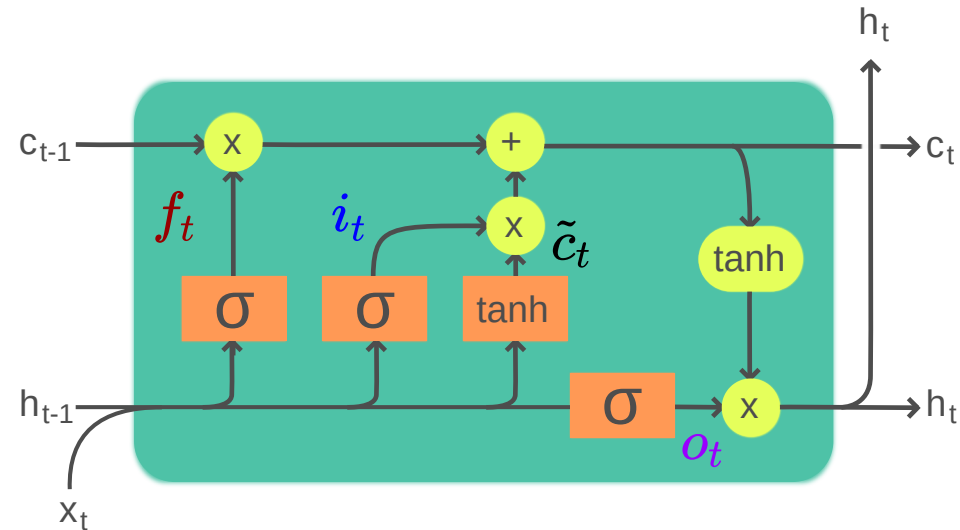
Images: Pascanu 2013

Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)

Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)



Legend:

Layer

Componentwise

Copy

Concatenate



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Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f)$$

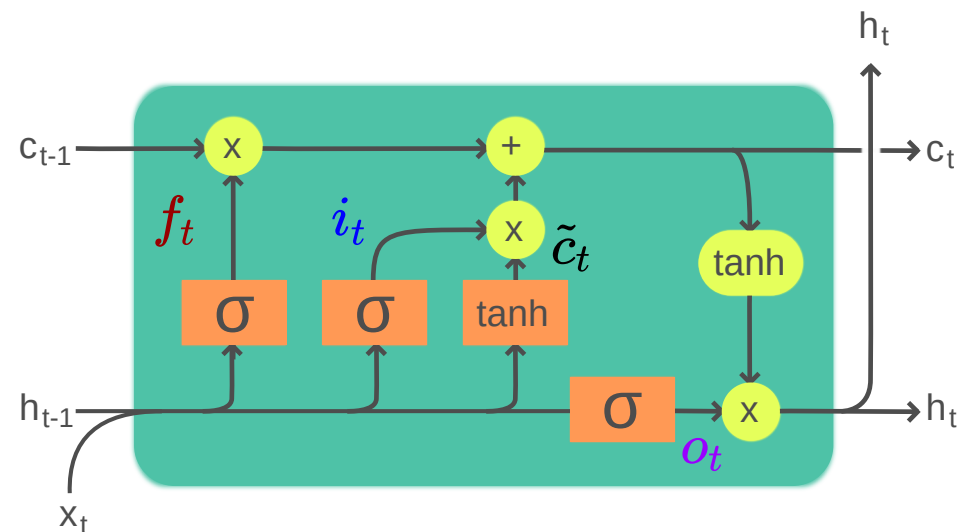
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:

Layer

Componentwise

Copy

Concatenate



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Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad \text{Forget gate}$$

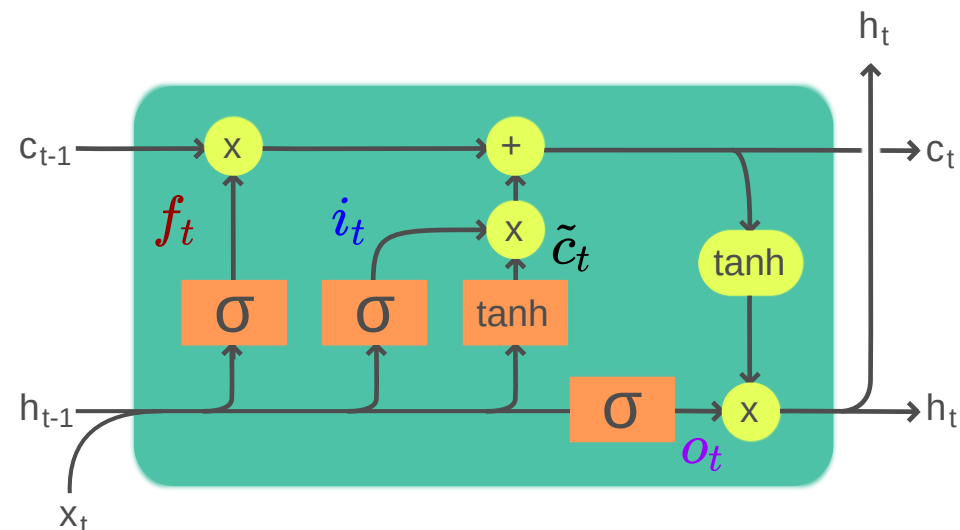
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:

Layer

Componentwise

Copy

Concatenate



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Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad \text{Forget gate}$$

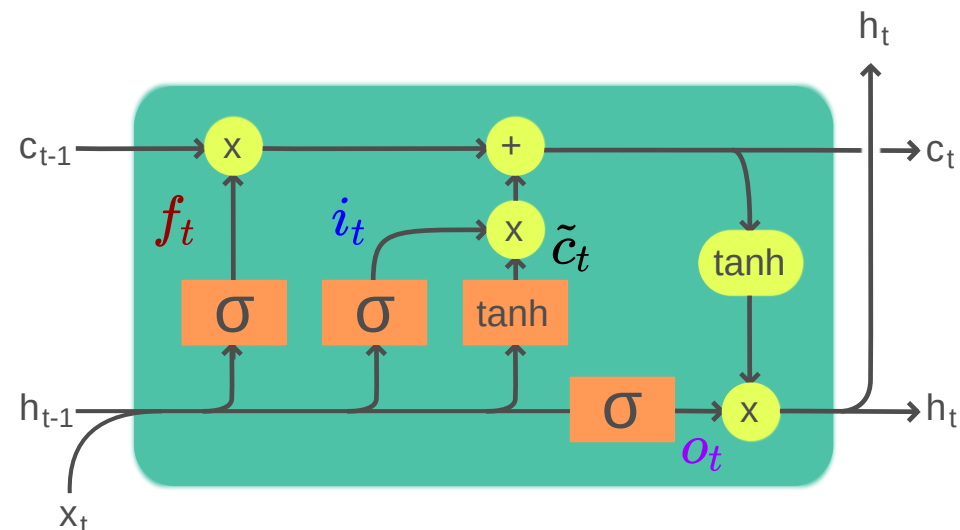
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \quad \text{Input Gate}$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:

Layer

Componentwise

Copy

Concatenate



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Approach 2: Recurrent Neural Network

Long Short-Term Memory (LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f)$ Forget gate

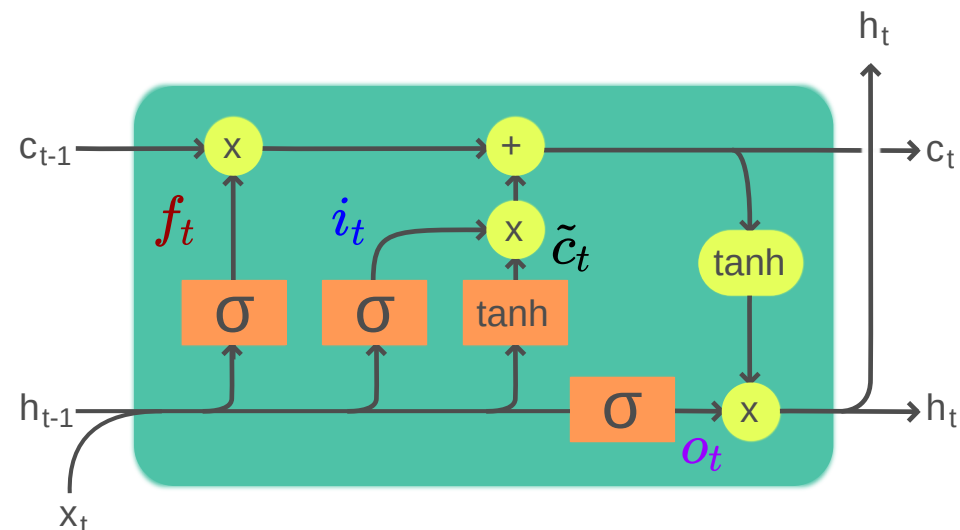
$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$ Input Gate

$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$ Output Gate

$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$

$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$

$h_t = o_t \odot \sigma_h(c_t)$



Legend:

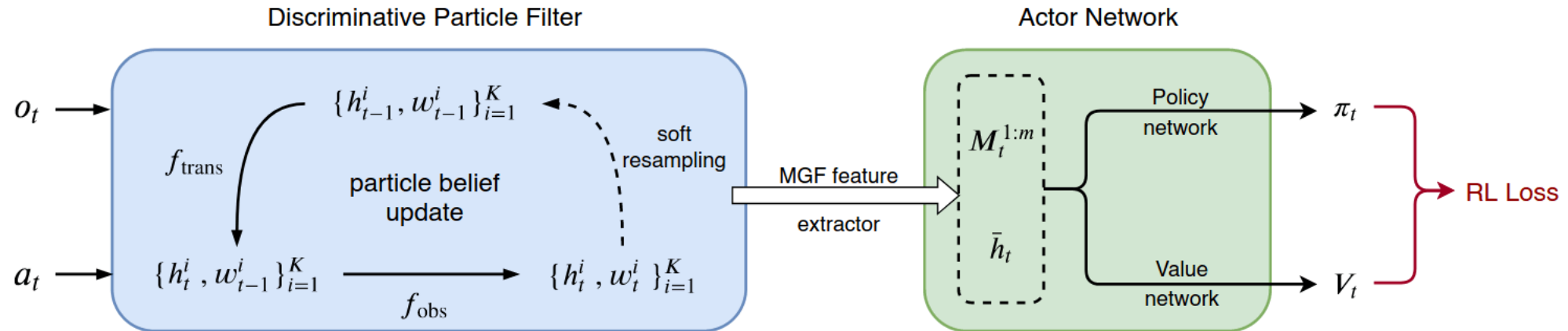
Layer Componentwise Copy Concatenate

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Approach 3: Particle Filters and MGF Features

Approach 3: Particle Filters and MGF Features



Published as a conference paper at ICLR 2020

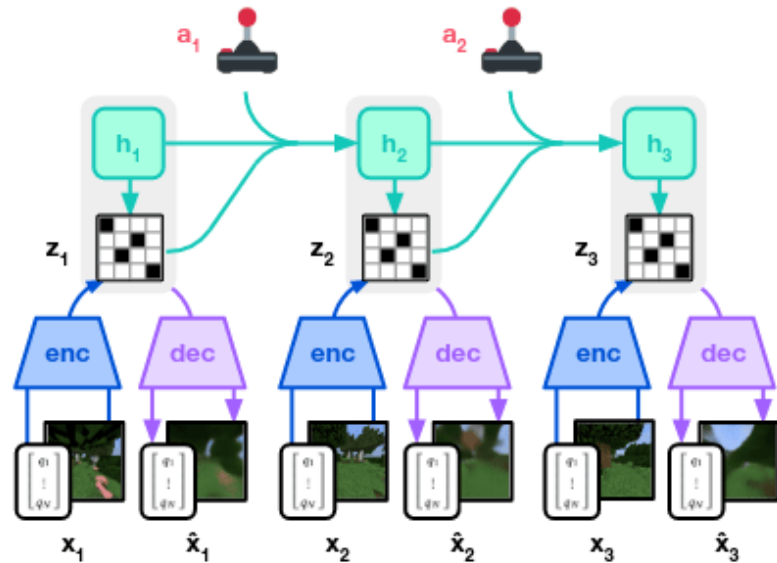
DISCRIMINATIVE PARTICLE FILTER REINFORCEMENT
LEARNING FOR COMPLEX PARTIAL OBSERVATIONS

Xiao Ma¹, Peter Karkus¹, David Hsu¹ and Wee Sun Lee¹, Nan Ye²

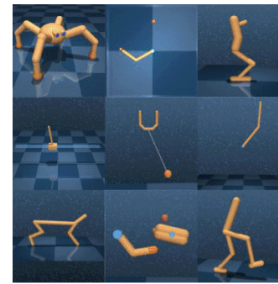
¹National University of Singapore, ²The University of Queensland

{xiao-ma, karkus, dyhsu, leews}@comp.nus.edu.sg, nan.ye@uq.edu.au

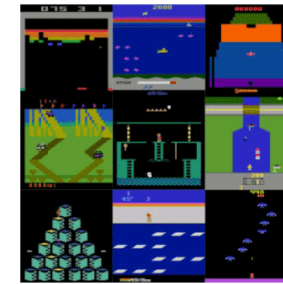
Approach 4: Latent Representations (e.g. World Models)



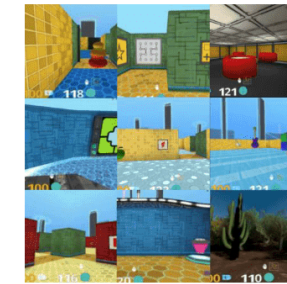
(a) World Model Learning



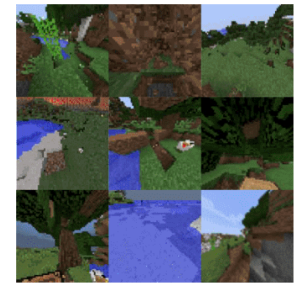
(a) Control Suite



(b) Atari



(c) DMLab



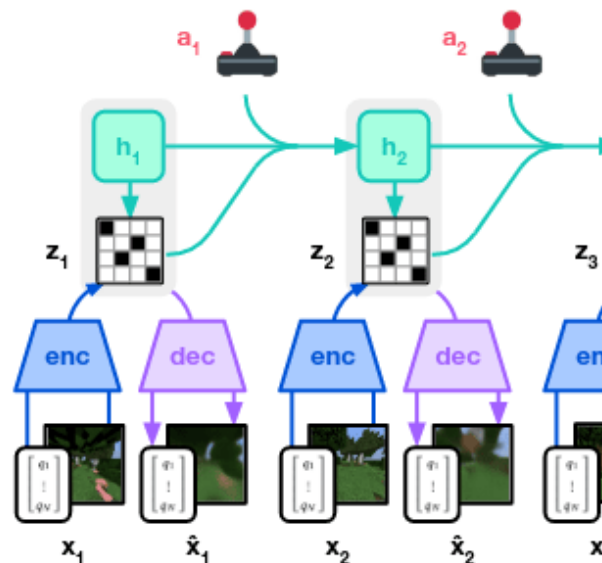
(d) Minecraft

Mastering Diverse Domains through World Models

Danijar Hafner,^{1,2} Jurgis Pasukonis,¹ Jimmy Ba,² Timothy Lillicrap¹

¹DeepMind ²University of Toronto

Approach



(a) World Model Learning

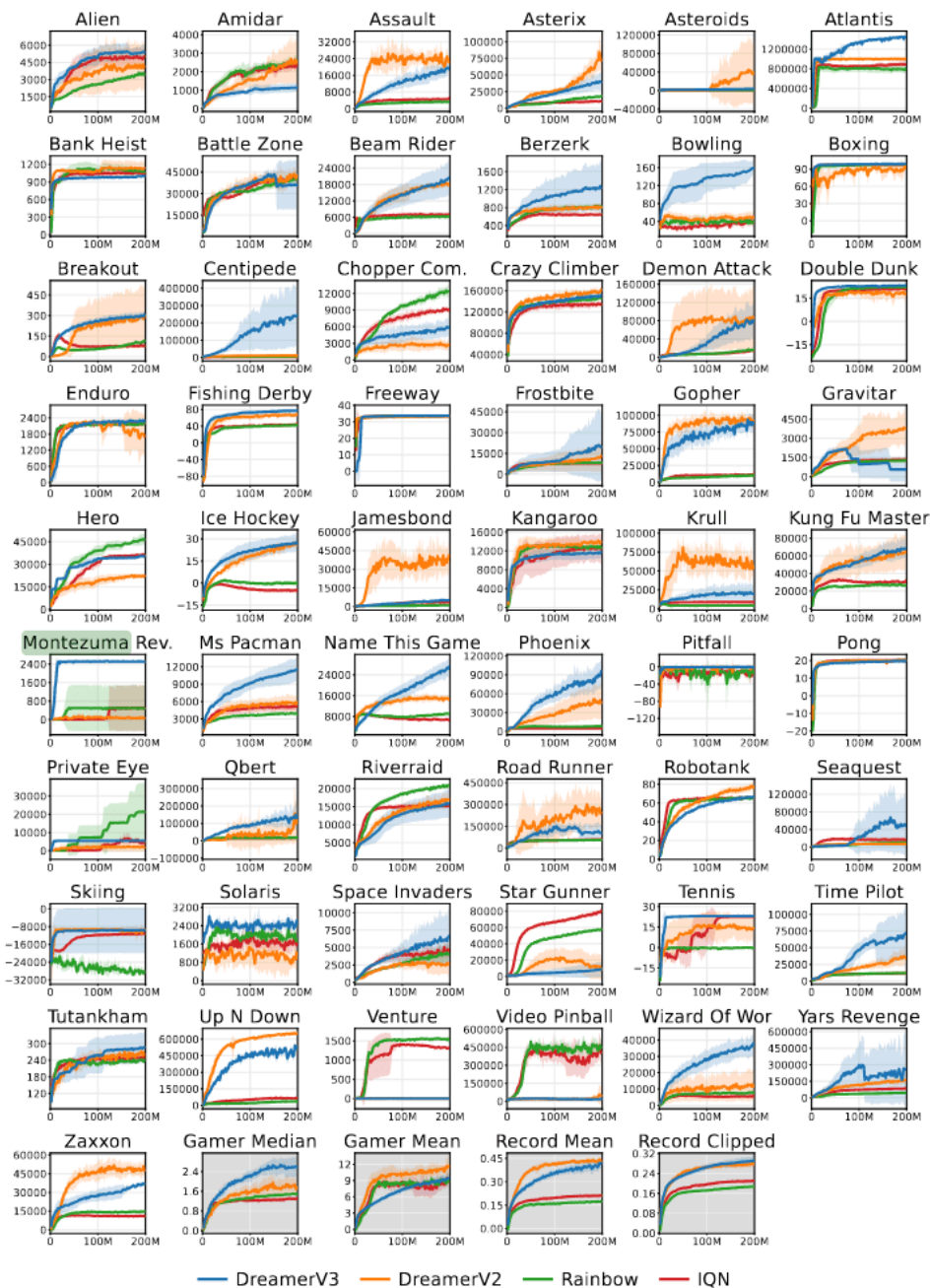
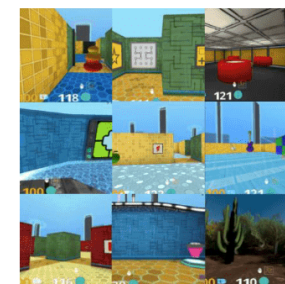


Figure U.1: Atari training curves with a budget of 200M frames.

Representations



(c) DMLab



(d) Minecraft

g Diverse Domains through World Models

Hafner,¹² Jurgis Pasukonis,¹ Jimmy Ba,² Timothy Lillicrap¹

¹DeepMind ²University of Toronto